



Detection of food additives based on an integrated self-injected metasurface microfluidic sensor

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Abstract: Advanced sensing equipment exhibits high sensitivity and reliability in detecting food additives, enabling the practical assessment of the safety of processed foods. Currently, chemical detection methods are commonly utilized for identifying food additives. However, these approaches tend to be intricate and time-consuming. In this study, we designed and fabricated an integrated terahertz microfluidic sensor, which achieves high sensitivity by incorporating a metasurface within the microfluidic chip. The metasurface comprises metal wires and split-ring resonators, with three optional sensing sites within the frequency domain of 0.1–1.2 THz, thereby enhancing the reliability of the sensor. Additionally, the use of a self-injection micropump improves the stability of the liquid flow rate, preventing experimental errors caused by manual injection. Utilizing this sensor, we conducted concentration sensing experiments on potassium sorbate and sodium benzoate solutions, successfully identifying sugar-containing and sugar-substituted beverages with high sensitivity and rapid sensing speed. The average sensitivity of the sensor is $152.8 \text{ GHz} \cdot \text{RIU}^{-1}$. The results of this study provide a feasible method for the development of microfluidic metasurface sensors.

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1. Introduction

Terahertz (THz) waves, falling within the frequency range of 0.1 to 10 THz [1–5], constitute a subset of electromagnetic radiation positioned between microwaves and far-infrared light. Their diverse applications encompass optical modulation [6], sensor detection [7], safety protection [8], medical applications, and research on dangerous goods [9,10], among others. THz waves possess unique characteristics, such as low energy levels [11], which prevent ionization damage during investigation and ensure the integrity of substances throughout measurement processes. Consequently, THz waves are particularly suitable for non-invasive sensing applications that require non-destructive analysis [12,13].

In a previous study, Toshihiko Kiwa et al. developed a THz chemical microscope (TCM) that leverages the fingerprint-like characteristics of THz waves. This innovative tool observed the binding between antigens and antibodies, with the detected THz pulse intensity varying significantly based on the binding conditions. This sensitivity highlights the potential of THz waves in a diverse range of analytical applications [14].

Despite numerous advantages, THz sensing encounters challenges. When studying substances using THz waves, measuring their transmission or absorption spectra is often necessary. However, the substantial absorption effect of water molecules on THz waves can mask solute absorption spectra, especially in aqueous solutions, compromising experimental outcomes [15]. Li et al. addressed the solution attenuation issue by proposing to reduce the interaction length between THz

waves and water molecules. Microfluidic technology emerged as a potential solution, utilizing capillary action to form a micron-sized liquid film [16,17]. This approach effectively decreases the interaction length between water and THz waves, thereby facilitating the acquisition of clearer solute absorption spectra [18]. However, relying solely on microfluidics for solution-sensing experiments may not be optimal. Under concentration gradients, the absorption spectra of substances often exhibit significant overlap, and variations in the intensity and frequency of absorption peaks can be unstable, rendering them unsuitable for sensing applications. Our controlled experiments in this study unequivocally highlight these challenges.

In comparison to other substances, metals exhibit a robust interaction with electromagnetic waves, particularly within the THz frequency range [19,20]. Consequently, artificially designed sub-wavelength metasurfaces have gained widespread utilization in THz sensing experiments. These metasurfaces exhibit one or more resonant absorption peaks within the THz frequency domain, maintaining a relatively consistent shape. This stability is minimally affected by the properties of the measured object, with only the peak intensity or frequency varying, thereby ensuring a high level of sensing stability. Furthermore, metasurfaces, crafted through parameter optimization, can yield sharp absorption peaks, facilitating high-sensitivity sensing. Recently, Li et al. designed a four-band metasurface sensor based on higher-order Fano resonance principles, demonstrating its efficacy in sensing hemoglobin solution with various thicknesses and concentrations [21]. This result underscores the viability of metasurface sensors in addressing the low-sensitivity issue associated with THz sensing.

Inspired by this, we integrated a microfluidic chip with a metasurface to overcome the attenuation challenges encountered by THz waves traversing solutions and the inherent instability of microfluidics. Furthermore, we coupled a specialized metasurface microfluidic sensor with a micropump to enhance experimental precision, effectively mitigating bubble formation from manual sampling. In this study, we conducted concentration gradient sensing experiments on sodium benzoate and potassium sorbate solutions of varying concentrations. Additionally, discrimination experiments were performed on sugary beverages and sugar-free beverages. The findings suggest that our designed metasurface microfluidic sensor can effectively differentiate between solutions of varying concentrations and discern solutions with distinct solutes. The results underscore the wide-ranging application potential of THz microfluidic sensors in the field of substance detection.

2. Experiment

2.1. Design of metasurface

The designed metasurface is presented in Fig. 1. The structure's center comprises metal wires that are perpendicular to each other. The substrate is divided into four square regions, each

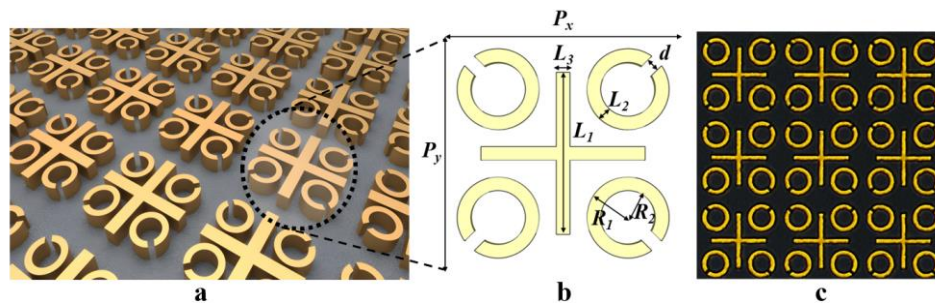


Fig. 1. (a) Schematic diagram of metasurface structure. (b) Schematic diagram of metasurface parameters. (c) Microscope image of the metasurface.

containing a split-ring resonator (SRR). The openings of the SRRs are arranged diagonally. The substrate is made of cycloolefin copolymer (COC) material, ensuring a THz transmission rate exceeding 90%. The metasurface is periodically arranged on the substrate. The unit length is $P_x = P_y = 180 \mu\text{m}$, with a repetition period of 100×100 . The metal wire length is $L_1 = 120 \mu\text{m}$, while the line width is $L_2 = L_3 = 10 \mu\text{m}$. The outer radius of the SRRs is $R_1 = 30 \mu\text{m}$, the inner radius is $R_2 = 20 \mu\text{m}$, and the opening length is $d = 10 \mu\text{m}$. The thickness of the gold layer is $h = 100 \text{ nm}$. The metasurface was manufactured using ultraviolet lithography. Further details regarding this process are provided in [Supplement 1](#).

2.2. Sensing principle

To elucidate the sensing principle of the metasurface microfluidic sensor comprehensively, we simulated the metasurface structure using the CST Studio Suite simulation software, as depicted in Fig. 2. Specifically, to enhance sensor stability, the pattern was intricately designed with central and axial symmetry, augmenting polarization insensitivity and ensuring adaptability to complex real-world scenarios. See [Supplement 1](#) for details about the simulation modeling.

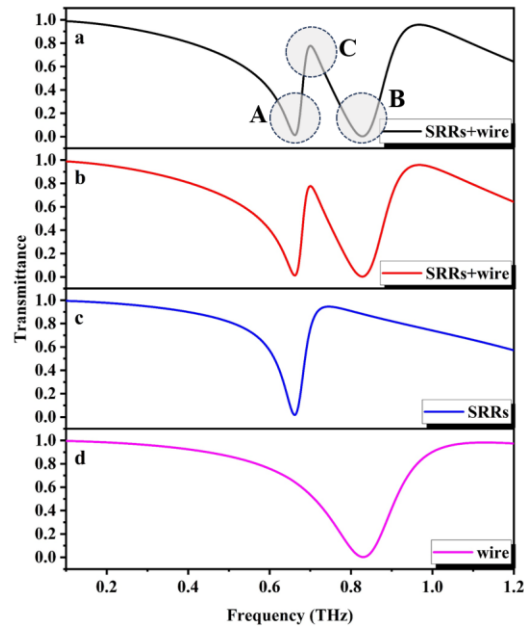


Fig. 2. Simulated transmission spectra with different structures: (a) X -polarized electric field with integrated structure. (b) Y -polarized electric field with integrated structure. Individual structure: (c) SRRs and (d) metal wires.

Figure 2(a) and 2(b) illustrate the transmission spectra for X -polarization and Y -polarization of the integrated structure, respectively, within the frequency range of 0.1–1.2 THz. These spectra exhibit two resonance absorption peaks, labeled A and B, and a transparent window, denoted as C, between them. Figure 2(c) and 2(d) provide further insights into the transmission spectra of the individual metal wire structure and split-ring resonators (SRRs) structure within the same frequency range. The resonance absorption peaks at A and B correspond to these distinct structural components. In the integrated structure, the metal wire exhibits a characteristic surface plasmon resonance (SPR) at 0.83 THz [22], originating from the vigorous oscillation of free electrons within the metal under the influence of an external electromagnetic field.

In contrast, SRRs produce a prominent LC resonance at 0.66 THz. Simulations were conducted to evaluate the surface electromagnetic enhancement effects of these two structures at their

respective resonant frequencies. Figure 3 illustrates the electric field distributions of the integrated and individual structures. Notably, the electric field of the metal wire is concentrated at both ends, whereas the electric field of the SRRs is focused at the slit.

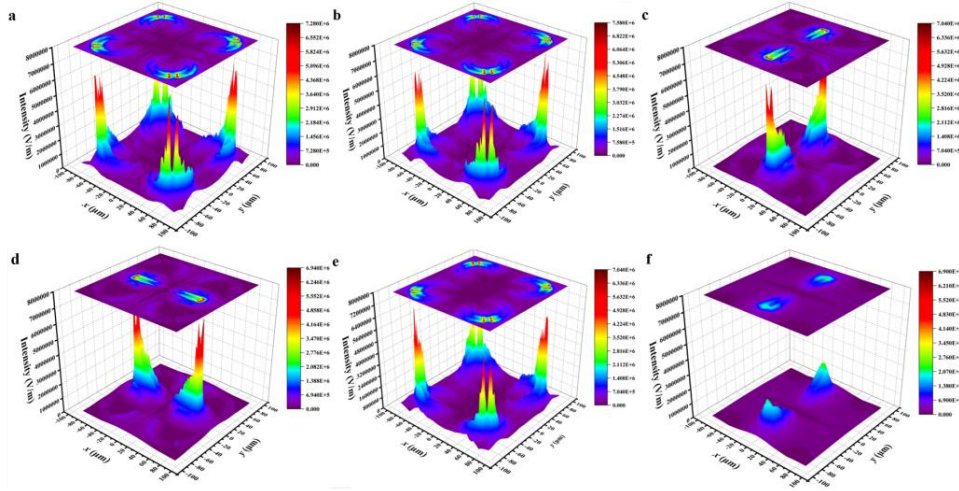


Fig. 3. Electric field mode of integrated structure: (a) Y -polarization electric field incidence, 0.66 THz. (b) X -polarization electric field incidence, 0.66 THz. (c) Y -polarization electric field incidence, 0.83 THz. (d) X -polarization electric field incidence, 0.83 THz. Electric field mode of individual structure: (e) Y -polarization electric field incidence, 0.68 THz. (f) Y -polarization electric field incidence, 0.80 THz.

Figure 3(a) and 3(c) depict the electric field distributions of the SRRs and metal wire when the electric field of the integrated structure is aligned with the Y direction. Conversely, Fig. 3(b) and 3(d) showcase the electric field distributions when the electric field polarization is aligned with the X direction. Figure 3(e) and 3(f) illustrate the surface electromagnetic enhancement effect of a single structure under exposure to a Y -polarized electric field. Our observations indicate that integrating these two structures effectively enhances the electromagnetic response capability of the metasurface, with a particularly notable effect observed in the metal wire structure.

The surface current maps presented in Fig. 4(a) to 4(d) reveal distinctive surface current flows for the two structures at their resonance absorption positions. Specifically, the cross-shaped metal wire primarily directs its current along the vertical metal wire under the Y -direction electric field. At the non-resonance frequency of 0.20 THz, the current is evenly distributed on both metal wires. In contrast, in SRRs, the surface current flows directionally at 0.68 THz. Moving on to Fig. 4(e) to 4(h), the simulated magnetic field images showcase the individual and integrated structures. The magnetic field distributions of the individual structures at their resonance frequencies are consistent with the surface current distributions. The metal wire structure generates a counterclockwise toroidal magnetic field under the Y -direction current, while the surface current on SRRs induces magnetic fields along the positive and negative Z axes.

When examining the magnetic field distribution of the integrated structure, the magnetic field generated by the surface current on the SRRs forms a closed loop near the metasurface at 0.66 THz. This closed-loop magnetic field excites a circumferential dipole along the X direction [23]. To represent this principle more intuitively, the magnetic field distribution on the cross-section is plotted to illustrate the toroidal magnetic field.

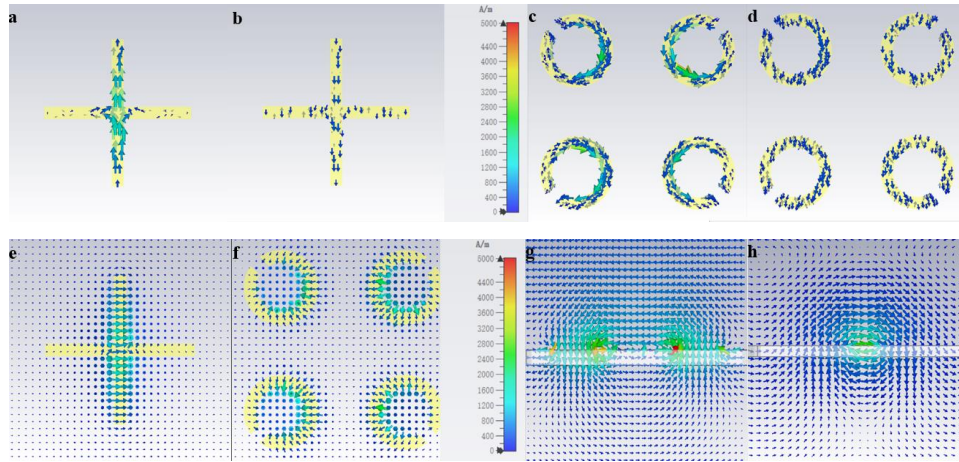


Fig. 4. Surface current distribution of metal wire structures: (a) 0.80 THz. (b) 0.20 THz. Surface current distribution of SRRs structures: (c) 0.68 THz. (d) 0.20 THz. (e) Magnetic field distribution of metal wire structure at 0.80 THz. (f) Magnetic field distribution of SRRs at 0.68 THz. Circular magnetic field in cross-section: (g) 0.66 THz, (h) 0.83 THz.

2.3. Parameter optimization

Metasurfaces, as artificial subwavelength structures, enable a wide range of sensing effects through precise fine-tuning of the sensor's structural parameters. To achieve optimal results within the limitations of the manufacturing process, we conducted detailed adjustments, utilizing the CST Studio Suite to simulate the effects of various metasurface parameters.

In Fig. 5(a) and 5(b), the metasurface is rotated at 0° , 45° , 135° , and 180° , and the transmission spectra are captured at each angle. The plotted spectra, along with schematics of the rotated structures, exhibit a distinct separation in transmission spectra for X -polarizations and Y -polarizations as the rotation angle increases. This separation is attributed to the loss of polarization symmetry in the split-ring resonators (SRRs) due to rotation, which results in reduced polarization insensitivity in the metasurface. At $\alpha = 180^\circ$, while the transmission spectra for the two polarizations are similar, the resonance absorption induced by the SRRs is nearly absent, leading to a loss of selectable sensing sites. Consequently, a rotation angle of 0° is considered the optimal choice.

Simultaneously, an attempt to align the openings of the four split-ring resonators (SRRs), facing in the positive direction of the Y axis, resulted in Fig. 5(c) displaying transmission spectra for Y -polarization incidence. In this configuration, the metasurface exhibits the electromagnetic induced transparency (EIT) effect, wherein the SRRs act as a “dark mode.” The resonance absorption of this mode is not directly excited by the Y -polarized electric field but rather by the magnetic field generated by the surface current in the metal wire. Figure 5(d) illustrates that, in separate structures, only the metal wire is excited by the Y -polarized electric field, while the SRRs are solely excited by the X -polarized field. However, in the integrated structure, resonance absorption of the SRRs occurs due to the dark mode being excited by the toroidal magnetic field generated by the current on the metal wire's surface. Consequently, this integrated structure demonstrates light-dark mode coupling, generating multiple resonances under incident Y -polarized electric fields [24]. However, this configuration significantly compromises polarization insensitivity, contrary to the initial intention.

Furthermore, we made adjustments to the line widths, varying the values of L_2 and L_3 to $10\ \mu\text{m}$, $12\ \mu\text{m}$, $14\ \mu\text{m}$, $16\ \mu\text{m}$, and $18\ \mu\text{m}$. Figure 5(e) displays simulated THz transmission spectra

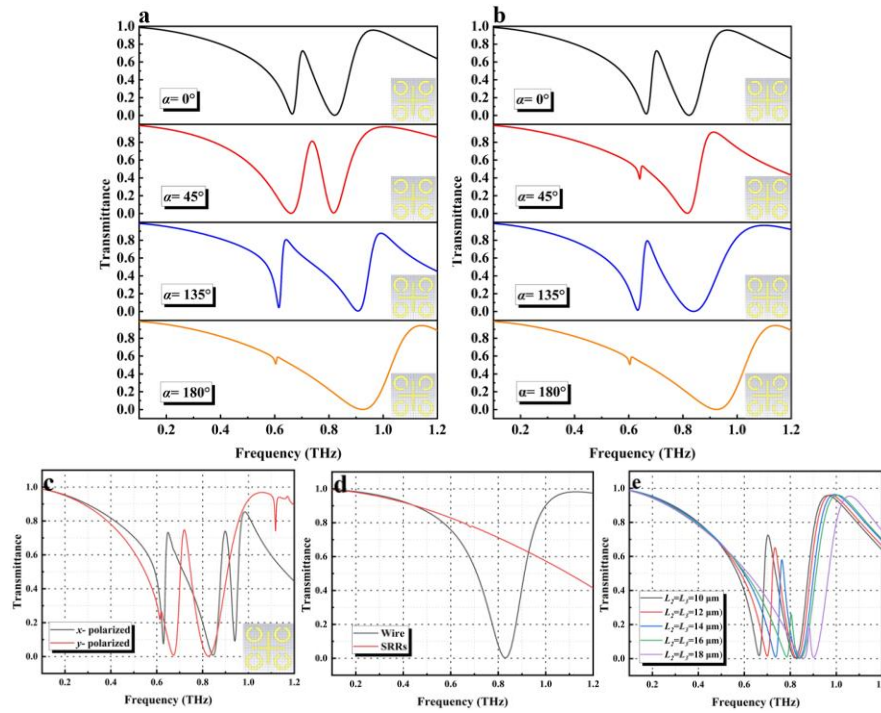


Fig. 5. (a) Y electric field polarization incident transmission spectra after SRRs are rotated 0°, 45°, 135°, 180°. (b) After SRRs is rotated 0°, 45°, 135°, 180°, X electric field polarization incident transmission spectra. (c) Transmission spectra of the integrated mechanism when the SRRs opening is upward and the polarization of the X and Y electric fields is incident. (d) Transmission spectra of both metal wire and SRRs structures when the SRRs opening is upward. (e) Transmission spectra with linewidths of 10 μm, 12 μm, 14 μm, 16 μm, 18 μm.

corresponding to these different line widths. Observably, as the line widths gradually increase, the spectra exhibit a noticeable blue shift, and the height of the transparent window decreases significantly. Additionally, the sensing sensitivity is reduced with the increase in the full width at half maximum (FWHM) of the resonance absorption peak. Therefore, for optimal sensing effects, the line width of $L_2 = L_3 = 10 \mu\text{m}$ is recommended as the final choice.

2.4. Sensor fabrication and experiment system

In this experiment, a self-injection metasurface microfluidic sensor, coupled with a micropump, was employed. The micropump ensured the consistent and stable pumping of the solution to be measured, thereby minimizing the formation of bubbles and significantly enhancing the accuracy of the experimental results. For further details, please refer to the “Sample Pumping In/Out” section in [Supplement 1](#).

The self-injection metasurface microfluidic sensor was produced as follows. Figure 6(a) provides a schematic of the sensor preparation. The sensor comprised two parts: a PMMA shell and a central portion made of COC. This design choice was motivated by the cost of COC material and the necessity to drill holes in the microfluidic chip for connection to an external micro-pump. By employing a detachable structure, we were able to prevent the flatness of the detection area from being compromised. Initially, a PMMA board with dimensions of $40 \times 40 \times 2$ mm was cut using a laser engraving machine to remove a 20×20 mm square. An L-shaped channel with a diameter of 0.7 mm was then drilled on its side using a drill bit. The left and

right holes were sealed with hot melt adhesive, and the detection area made of COC material was subsequently embedded within it. Deionized water was then pumped into the sensor to test its sealing performance.

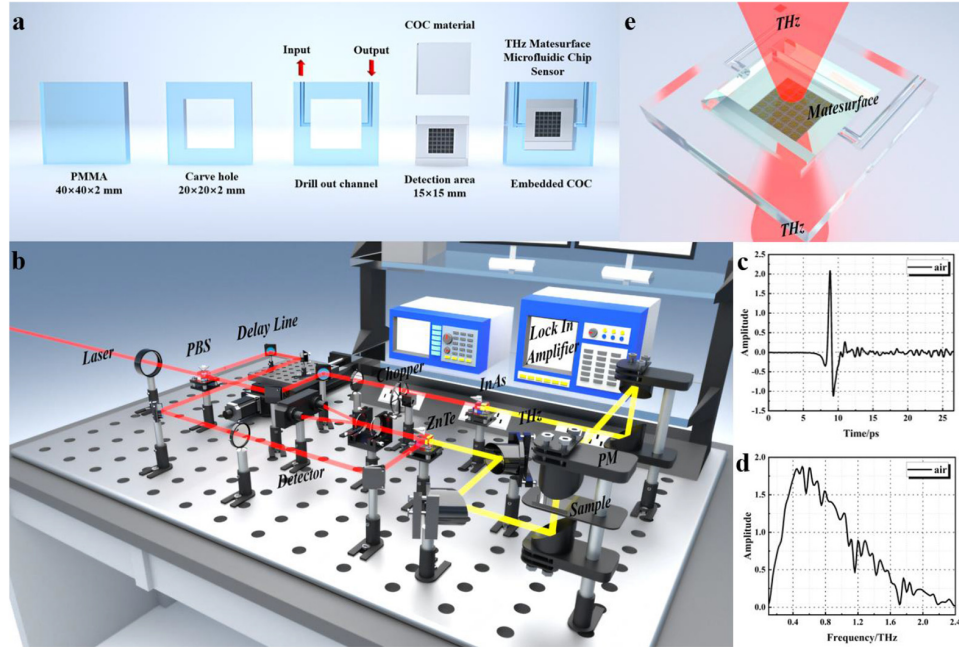


Fig. 6. (a) Schematic diagram of sensor manufacturing process. (b) Schematic diagram of experimental system. (c) Time domain spectrum of the system. (d) Frequency domain spectrum of the system. (e) Sensor working diagram.

In particular, the sample chamber was constructed using two COC materials, each with dimensions of 20×20 mm and a thickness of 1 mm. A high-precision milling machine was employed to mill the center region of these materials, creating a detection area of 20×15 mm. It is noteworthy that both COC undergo milling, resulting in a detection area thickness that is equivalent to the combined thicknesses of the two COC sample chambers, totaling $50 \mu\text{m}$. Consequently, the milling depth on each COC was $25 \mu\text{m}$. Subsequently, a metasurface with a thickness of 100 nm was fabricated on one of the COC in the detection area. The two COC layers were then bonded using oxygen plasma bonding and subsequently installed into the PMMA holes, with the edges sealed using hot glue. For a detailed description of the preparation process of the COC detection area, please refer to [Supplement 1](#).

As shown in Fig. 6(b), the experimental THz-TDS system mainly consists of a femtosecond laser, a time delay control system, a THz wave generation device, and a detection device, with an effective bandwidth range of 0.1-2.4 THz. The laser used is an 800 nm femtosecond laser (output power: 130 mW, center wavelength: 800 nm, pulse repetition frequency: 100 MHz, pulse width: 75 fs). The laser beam is divided into pump light and probe light by a polarizing beam splitter (PBS). The pump light passes through a chopper and a delay line, generating THz waves via the optical rectification effect. These THz waves are then focused by an off-axis parabolic mirror, illuminating the sample and carrying the sample signal. Both the THz wave and the probe light enter a ZnTe crystal from different directions, undergoing modulation by the electro-optic effect, and are subsequently detected by a differential detector. The collected data are finally processed by a computer. The time-domain and frequency-domain diagrams of the optical path of the system are presented in Fig. 6(c) and (d), respectively. In our experiment, the microfluidic sensor

incorporating a metasurface is connected to a micro-pump, facilitating the uniform pumping of the liquid to be tested. Figure 6(e) illustrates a schematic diagram of the sensor's working principle.

3. Results and discussion

3.1. Concentration sensing of potassium sorbate and sodium benzoate solutions

Potassium sorbate, the potassium salt of sorbic acid, is highly water-soluble and commonly utilized as a food preservative, with an established acceptable daily intake level for humans set at 25 mg/kg. However, potassium sorbate can exhibit toxicity under certain conditions, particularly at high concentrations or when combined with nitrite. Therefore, stringent dosage limitations must be adhered to when using potassium sorbate as an additive. In this regard, we conducted sensing experiments on potassium sorbate solutions using the aforementioned self-injection metasurface microfluidic sensor.

A micro pump facilitated the consistent injection of potassium sorbate solution into the sensor, thereby eliminating potential interference from bubbles during the experiment. Six different concentrations of potassium sorbate solutions were precisely prepared, ranging from 1 mg/ml to 6 mg/ml with incremental concentration gradients. Additionally, deionized water was pumped into the sensor as a control group during the experiments. After each measurement, the channels and detection areas were thoroughly cleaned with deionized water, followed by air drying. The THz spectra were then measured to obtain the metasurface sensor's resonance response spectra under varying solution concentrations.

Figure 7(a) displays the THz transmission spectra of potassium sorbate solutions at six different concentrations, exhibiting a redshift in the spectra as the concentration increases, along with fluctuations in the resonance intensity. The frequency shift (Δf) corresponding to each sensing site under varying concentrations was defined. Figure 7(c) illustrates the variation curve of Δf with solution concentrations at three sensing sites, showing a roughly linear trend. This result indicates that the change in solution concentration exhibits a reasonably good linear relationship with Δf . The experimental sensitivity (β) of the sensor for detecting potassium sorbate solutions is defined as $\beta = \Delta f / \Delta \text{Concentration}$. The values for the three sensing sites ($\beta_1 = 24.7$, $\beta_2 = 25.6$, $\beta_3 = 23.8$, Units: GHz/(mg·ml⁻¹)) indicate significant sensing capabilities.

Furthermore, a conventional microfluidic chip (as depicted in Fig. 7(i)) was used as a control experiment to emphasize the importance of incorporating a metasurface. To highlight the influence of water molecules on the solution experiment, additional THz absorption spectra of potassium sorbate solid tablets were included. The absorption coefficient $\alpha(\omega)$ of potassium sorbate at various concentrations was calculated using the formula:

$$\alpha(\omega) = \frac{2\kappa(\omega)\omega}{c} = \frac{2}{d} \ln \frac{4n(\omega)}{[n(\omega) + 1]^2} \quad (1)$$

$$n(\omega) = \frac{\varphi(\omega)c}{\omega h} + 1 \quad (2)$$

In the given formula, where c represents the speed of light, h denotes the thickness of the sample, ω signifies the angular frequency of the signal, $A(\omega)$ and $\varphi(\omega)$ stand for the amplitude and phase ratio of the Fourier transform spectrum of the sample signal and reference signal, and $\kappa(\omega)$ and $\alpha(\omega)$ represent the extinction coefficient and absorption coefficient of the sample, respectively. Within the frequency range of 0.1-2.4 THz, although characteristic absorption of potassium sorbate is observable, it lacks linearity, and the spectral lines' stability is suboptimal, posing challenges in identifying fixed sensing sites. Simultaneously, microfluidics falls short of completely mitigating the impact of water molecules on sensor experiments. Comparing the THz absorption spectra of solid potassium sorbate in Fig. 7(i), it is evident that water as a solvent

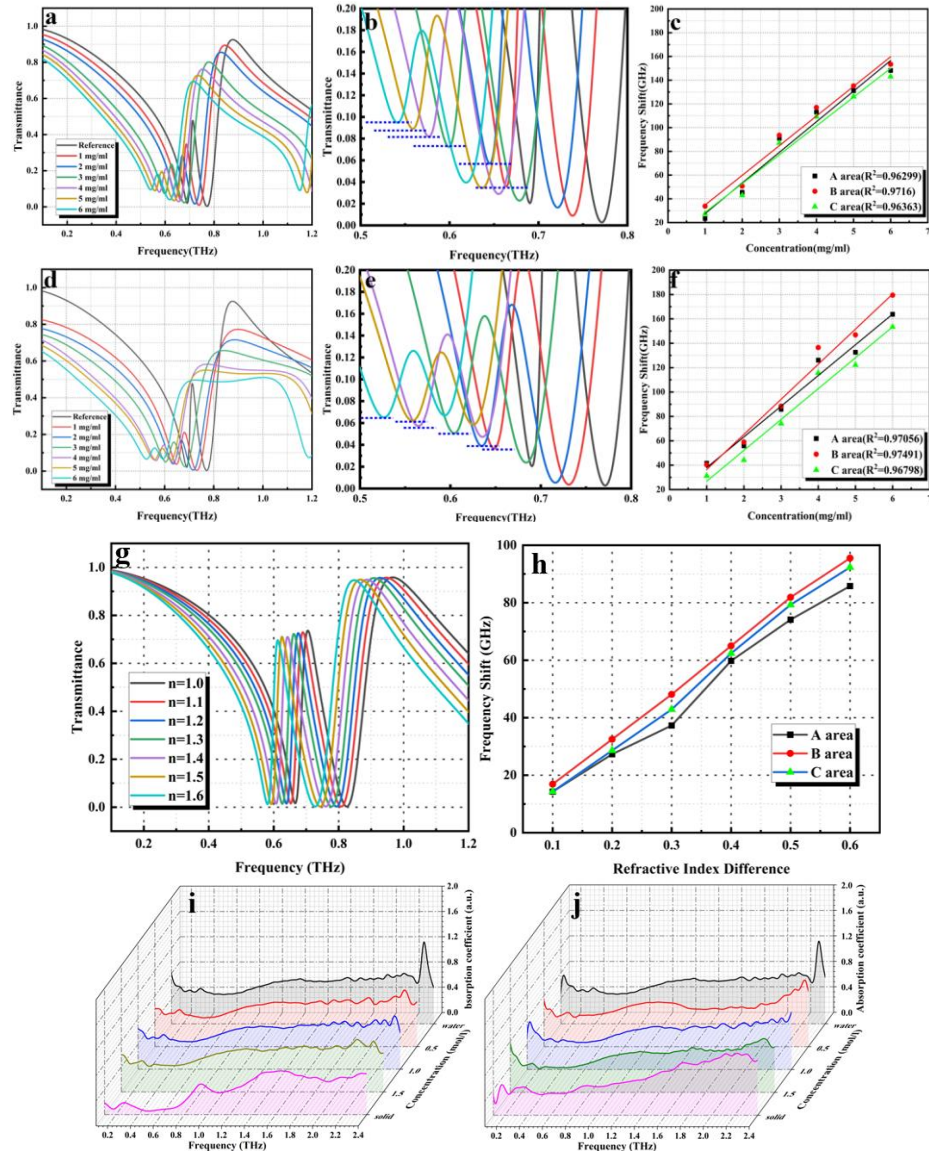


Fig. 7. (a) Concentration sensing spectra of sensors for the detection of potassium sorbate solutions. (b) Local magnification of potassium sorbate sensing spectra. (c) Frequency shift of potassium sorbate solutions concentration sensing spectra. (d) Concentration sensing spectra of sensors for detecting sodium benzoate solutions. (e) Local magnification of sodium benzoate sensing spectra. (f) Frequency shift of sodium benzoate solutions concentration sensing spectra. (g) Numerical simulation of concentration sensing spectra. (h) Frequency shift simulation of concentration sensing spectra. (i) Concentration sensing spectra of potassium sorbate solutions detected by ordinary microfluidic chip. (j) Concentration sensing spectra of sodium benzoate solutions detected by ordinary microfluidic chip.

not only conceals a portion of the THz absorption peak of the solute but also complicates the interpretation of the THz resonance absorption peak of the composite system. Hence, ordinary microfluidics proves unsuitable for solution-sensing detection experiments.

Similarly, sodium benzoate, a water-soluble broad-spectrum antimicrobial reagent commonly used in food preservation, underwent sensing experiments utilizing the metasurface microfluidic sensor. The persistent influence of water molecules on sensing experiments is evident in Fig. 7(j). The gradient experiments using ordinary microfluidics did not exhibit significant differences compared to solid-state experiments. Figure 7(d) illustrates the THz transmission spectra measured using the metasurface microfluidic sensor, with sodium benzoate solution concentrations ranging from 1 mg/ml to 6 mg/ml. As the concentration increases, the spectra undergo a corresponding redshift, and the average total frequency shift is 165.5 GHz. Figure 7(f) depicts the change curve of the frequency shift at three sensing sites with respect to the concentrations of sodium benzoate solution. All curves exhibit good linearity. The experimental sensitivity γ of the sensor for detecting sodium benzoate solution is defined as $\gamma = \Delta f / \Delta \text{Concentration}$. The γ values at the three sensing sites ($\gamma_1 = 27.3$, $\gamma_2 = 29.9$, and $\gamma_3 = 26.5$, Units: GHz/(mg·ml⁻¹)) highlight significant sensing effects.

To more clearly illustrate the distinct effect of the sensor on the aforementioned two solutions, Fig. 7(g) and Fig. 7(h) provide a localized magnification of the resonance absorption peaks of the sensing spectral lines for the two solutions. It can be observed that, at the same concentration, the absorption peak frequency shift of sodium benzoate solutions is significant, and the transmittance is low. This characteristic can serve as a basis for distinguishing between the two solutions.

In particular, as the concentration of the two solutions increases, the depth of the THz resonance diminishes, and the transmittance of the transparent window decreases. This phenomenon is likely due to the enhanced concentration of ions in the solution with the addition of the sample, thereby intensifying the scattering effect on THz waves.

Figure 7(g) depicts the simulation of transmission spectra for concentration sensing experiments, incorporating refractive indices of $n = 1.0, 1.1, 1.2, 1.3, 1.4, 1.5$, and 1.6 . In the simulation, these refractive indices served as variables for parametric scanning. The theoretical sensitivity of the sensor, denoted as $\tau = \Delta f / \Delta n$, resulted in calculated τ values at three sensor sites: 162.5, 142.5, and 153.3 (Units: GHz·RIU⁻¹), respectively.

The combination of experiments and simulations underscores the designed metasurface microfluidic sensor's ability to discern different concentrations of solutions. This ability is clearly demonstrated by the corresponding frequency shift observed in response to concentration changes. The frequency shift displays regularity and stability, closely aligning with the requirements of high-speed and high-sensitivity sensors.

3.2. Identification of sugar-free drinks and sugary drinks

Sugar alternatives, such as erythritol and xylitol, are commonly used in sugar-free drinks, functional beverages, and solid foods, offering a safe alternative for individuals with diabetes or hyperglycemia within specific dietary limitations [25]. Despite their advantages, the high cost of sugar alternatives has led to instances of sugar-containing beverages being falsely labeled as sugar-free, posing significant safety concerns. Therefore, it is crucial to develop a sensor capable of quickly distinguishing between these two types of solutions.

Glucose and erythritol were configured in gradient solutions ranging from 1 mg/ml to 6 mg/ml, respectively. Glucose solutions represent sugary beverages, while erythritol solutions represent sugar-free beverages. A micropump delivered the self-made solutions into the sensor to measure the THz transmission spectra, and detection experiments were conducted on the self-made solutions, as depicted in Fig. 8(a) and Fig. 8(b). The three sensing sites exhibited significant frequency shifts. By comparing the spectra of the two solutions, it can be observed that the corresponding frequency shift of erythritol solutions is larger, suggesting that erythritol solutions have a higher dielectric constant at the same concentration. To verify this speculation, the real and imaginary parts of the dielectric constants of the two solutions were measured, and the results are presented in Fig. S5. Furthermore, the shape of erythritol's spectral line (i.e., dip depth and

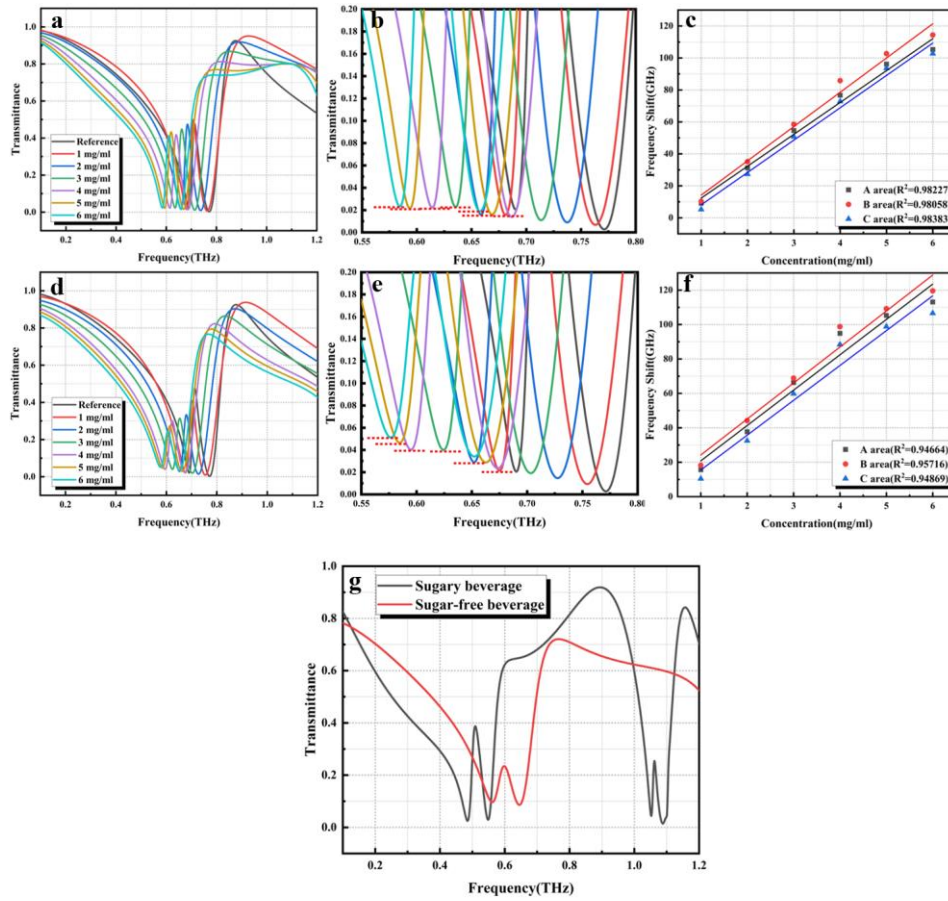


Fig. 8. (a) Glucose solutions sensing lines. (b) Local magnification of the sensor lines of glucose solutions. (c) Spectral line frequency shift of glucose solutions. (d) Erythritol solutions sensing lines. (e) Local magnification of the sensor lines of erythritol solutions. (f) Frequency shift of the sensor line in erythritol solutions. (g) Sensing lines for sugary beverages (black line) and sugar-free beverages (red line).

transparent window height) undergoes more significant changes under the concentration gradient. The relationship between the dielectric constant and the conductivity of the solution is described in [26]:

$$\varepsilon = \varepsilon' + i\varepsilon'' \quad (3)$$

$$\sigma = \sigma' + i\sigma'' \quad (4)$$

$$\sigma' = \frac{\omega\varepsilon''}{4\pi} \quad (5)$$

$$\sigma'' = (1 - \varepsilon')\frac{\omega}{4\pi} \quad (6)$$

where ε' is the real-part of dielectric constant, ε'' is the imaginary-part of dielectric constant, σ' is the real-part of conductivity, σ'' is the imaginary-part of conductivity, and ω is the frequency of incident field. It can be seen that σ is positively correlated with ε'' . When the frequency of the incident field increases, the electro polarization process of the solution cannot follow the change of the external field [27], so that the dielectric constant of the solution decreases in both real and

imaginary parts [28]. When different concentrations of solutions are at the same frequency, the concentration of ions is different. As the ion concentration increases, the conductivity of the solution correspondingly rises, and the dielectric constant of the solution exhibits a concurrent augmentation with the elevation of concentration. At the same concentration, erythritol solutions have a higher ion density than glucose solutions, which makes erythritol solutions have a stronger scattering effect on THz, resulting in a greater change in the shape of the spectral lines. In addition, the ionization of erythritol and glucose in the solutions is weaker than the above two salt solutions, so the spectral line changes of the two are not as obvious as the salt solution. Therefore, the frequency shift and shape change of the sensing spectral lines are the result of the joint action of dielectric constant and electrical conductivity.

In particular, THz solution discrimination experiments were conducted on two common sugary drinks (regular cola) and sugar-free drinks (diet cola) purchased from the market, as depicted in Fig. 8(g). The sensor demonstrates a high level of discrimination between the two substances. The observed differences in the sensing spectral lines between the commercially available beverages and the self-prepared standard solutions can be attributed to two primary factors. Firstly, the beverages may contain a small quantity of additional substances, apart from glucose or erythritol. Secondly, the concentration of erythritol required to achieve a comparable level of sweetness in a commercially available beverage may be lower than the concentration of glucose, thus contributing to the observed spectral variations.

4. Conclusion

To address the issue of low detection sensitivity in conventional microfluidic sensors, we introduce a THz microfluidic sensor that integrates a metasurface. This design incorporates a composite metasurface comprising split-ring resonators and metal wires, along with microfluidic technology. The sensor is coupled with a micropump to enhance precision and mitigate errors associated with manual sampling. The sensor effectively detects the concentrations of potassium sorbate and sodium benzoate solutions, identifies solutions of erythritol and glucose, and discriminates between common sugary and artificial sweetener beverages readily available in the market.

This sensor is compact, highly integrated, offers rapid detection, and is exceptionally sensitive. It is ideally suited for detecting trace amounts of liquid and facilitates the prompt acquisition of the approximate content of the detection object. Moreover, it presents a viable solution for the practical application of THz detection technology.

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Data availability. All relevant data are within the paper and its supplementary material.

Supplemental document. See [Supplement 1](#) for supporting content.

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